

Skills Summary

Second and Higher Derivatives

Use this reference while completing your worksheet or when studying.

Skill 1 — Repeated Differentiation

Notation. f' , f'' , f''' , then $f^{(4)}$, $f^{(5)}$, $\dots$

Each one is the derivative of the one before it: $f'' = (f')'$ and $f''' = (f'')'$. Differentiate once, simplify, differentiate the result.

Example: Find $g''(x)$ for $g(x) = x^5 - 4x^3 + 2x$.

Step 1. The order asked for is two, so this is the power rule applied twice — no new tool is needed.

Step 2. $g'(x) = 5x^4 - 12x^2 + 2$, and differentiating that gives $20x^3 - 24x$.

$$\Rightarrow g''(x) = 20x^3 - 24x$$

Try it: Find $h''(x)$ for $h(x) = x^4 - 6x^2 + 9$.

$$\text{check: } h''(x) = (x^4 - 6x^2 + 9)'' = 4x^3 - 12x$$

Skill 2 — Acceleration from a Position Model

Rule. If $s(t)$ is position, then $v(t) = s'(t)$ is velocity and $a(t) = s''(t)$ is acceleration.

Units divide once per derivative: metres $\rightarrow$ metres per second $\rightarrow$ metres per second squared.

Example: $s(t) = t^3 - 6t^2 + 5t$ metres. Find $a(t)$ and $a(4)$.

Step 1. Acceleration is the second derivative of position, so differentiate twice and only then substitute the time.

Step 2. $s'(t) = 3t^2 - 12t + 5$, so $a(t) = s''(t) = 6t - 12$; at $t = 4$, $a(4) = 24 - 12$.

$$\Rightarrow a(t) = 6t - 12 \quad \text{and} \quad a(4) = 12 \text{ metres per second squared.}$$

Try it: $s(t) = 2t^3 - 9t^2 + 4t$ metres. Find $a(t)$ and $a(3)$.

$$\text{check: } a(3) = (2t^3 - 9t^2 + 4t)'' = 6t - 18 = 18 - 18 = 0$$

Skill 3 — Second Derivative of an Implicit Curve

Recipe. Differentiate the equation term by term, remembering $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$. Solve for $\frac{dy}{dx}$, differentiate again, then replace any surviving $\frac{dy}{dx}$ by the expression you already found.

Example: Find $\frac{d^2y}{dx^2}$ for the curve $xy = 6$.

Step 1. The curve solves cleanly for y , so the fastest honest route is to write y explicitly and differentiate twice; implicit differentiation gives the same answer.

Step 2. $y = 6x^{-1}$, so $y' = -6x^{-2}$ and $y'' = 12x^{-3}$.

$$\Rightarrow y'' = \frac{12}{x^3}$$

Try it: Find $\frac{d^2y}{dx^2}$ for the curve $xy = 10$.

check: $\frac{d^2x}{dy^2} = \frac{1}{y''}$